

Document Q&A; Bot using RAG - Complete Assignment Guide

This document contains the complete implementation plan for the AI Engineering Internship assignment.

Project Structure:

document-qa-bot/

- .env
- requirements.txt
- README.md
- data/
- db/
- src/
- config.py
- ingest.py
- query.py
- main.py

Core Features:

- PDF and DOCX ingestion
- Text chunking with overlap
- ChromaDB persistent storage
- Gemini embeddings
- Retrieval-Augmented Generation (RAG)
- Grounded answers with citations
- Streamlit web interface

Execution Steps:

1. Create virtual environment:

```
python -m venv venv
```

2. Activate environment:

Windows: `venv\Scripts\activate`

3. Install dependencies:

```
pip install -r requirements.txt
```

4. Create .env file:

```
GEMINI_API_KEY=your_api_key
```

5. Add documents to data/ folder.

6. Run ingestion:

```
python ingest.py
```

7. Launch application:

```
streamlit run main.py
```

Human intervention is required only for:

- Adding documents
- Adding API key
- Re-running ingestion when documents change

For full source code, refer to the chat response.

Gemini API Key Setup

1. Visit Google AI Studio: <https://aistudio.google.com/>
2. Sign in with your Google account.
3. Click 'Get API key'.
4. Click 'Create API key'.
5. Copy the generated key.
6. Paste it into your .env file:
`GEMINI_API_KEY=your_key`